

Why live opportunities come in thin — a coverage audit

Six sources audited against the code, not assumptions. The finding is systemic but single-rooted: we ingest the *listing* of every opportunity and almost never the *detail* — and the ingestion model has no slots for the fields that matter.

THE FINDING IN ONE PARAGRAPH

Listing-deep, not detail-deep

Every live-opportunity source captures the **search-results row** (title, buyer, close date, link) and stops there. The rich data — scope, discipline, estimated value, buyer contact, and the RFP documents — lives on each opportunity's **detail page** or deeper in the **feed payload**, and almost nothing pulls it. The database proves it: across all live sources, **Discipline, EstimatedValue, and BuyerContactEmail are populated on essentially zero rows**. This is not ten separate bugs — it's one architecture that was built to catch opportunities, never wired to *understand* them.

Two layers, and they stack

LAYER 1 - THE INGESTION MODEL HAS NO SLOTS

`OpportunityCandidate` (the only object a source can emit) has **no field for Discipline, no BuyerContactEmail, and no Documents/attachments**. So even a source that exposes them has nowhere to put them — everything not modelled is serialized into `RawJson` and dropped from queryable columns. `OpportunityCandidate.cs:16-56`

LAYER 2 - THE PERSISTENCE STEP HARDCODES THE MISS

`IngestionService.BuildOpportunity` stamps **Discipline = Unknown on every opportunity** and never assigns a buyer contact. `EstimatedValue` is the *only* enrichment column wired to a source value — so even where a scraper reads the commodity codes (BC Bid literally reads "81101505 Structural engineering"), they're dumped into free-text and the discipline stays Unknown. `IngestionService.cs:578-595`

AND THERE IS NO OPPORTUNITY-LEVEL ENRICHMENT LAYER TO CATCH IT LATER

The one enrichment spine that exists — `EnrichmentDispatcher` — enriches canonical **organizations** (company profiles), not opportunities. The only code that ever opens an opportunity's detail page is (a) the **historical/closed-bid** pipeline, (b) **plan-takers** for BC Bid + APC, and (c) two **unwired one-off probe tools** that already log in and read detail pages but were never connected to production.

PER-SOURCE RESULTS

Where the thinness is — and whether it's fixable

SOURCE	VISITS DETAIL ON LIVE OPPS?	WHAT'S DROPPED (AND AVAILABLE)
BC Bid (266+113 opps)	No — listing grid only	scope, commodity/discipline codes, value, official contact, RFP docs, contract term
Bids&Tenders (~27 tenants)	No — never opens the tender	scope, category, cont the RFP PDFs . Plus Interior Health isn't ev configured tenant
MERX-DCC (60, incl. CFHA)	No — public listing row only	scope, contact, documents, plan- holders , even the clo: <i>time</i>
Bonfire (~35 tenants: UBC, Fraser Health, PHSA, Island Health, VCH)	No — RSS feed only	full RFP scope, categ and documents behin each <code>/opportunities/{id}</code>
CanadaBuys (national)	n/a — CSV feed	contactInfoEmail/ph GSIN/UNSPSC codes city, attachment URI all present per row, dropped to RawJson
SAM.gov (US)	n/a — API	pointOfContact.email, resourceLinks, naicsCode→discipline description
APC (Alberta, 156)	Yes — but reads only plan- takers	scope, contact, categ docs — read <i>past</i> on already-loaded page
CivicInfo BC	No — listing/RSS	detail pages exist, unvisited
Major Projects Inventories (AB/BC/CA ×4)	n/a — feeds	rich value/owner/stag captured, but emit ze opportunity rows — siloeed in a separate ta
BC Gov News RSS · BD Alerts email	No	nothing meaningful to pull
GovCanada feeds	n/a	it's an <i>awards</i> feed → writes to OpportunityAward, no live opps



Only two sources are legitimately headline-only (news RSS, alert emails). Everything else is dropping data that is provably present at the source.

THE GOOD NEWS

The fix pattern already exists and runs in production

APC IS THE TEMPLATE

APC gets plan-takers because it has a **separate, decoupled, idempotent enrichment job** (`ApcInterestEnrichmentJob` + `ApcInterestExtractor`): it selects live opps lacking enrichment, builds the detail URL from the stable reference, renders the page, extracts structured rows, and upserts — with per-item try/catch so it's resume-friendly. It **already loads the full detail DOM** and then extracts only supplier cards. The mechanism every other source is missing is already built, proven, and running — it just needs to (a) exist for the other sources and (b) harvest the full field set on the pass it already makes.

PROPOSED FIX — NOT STARTED, FOR YOUR GO-AHEAD

A four-rung ladder, cheapest first

TIER 0 — FREE WINS, DATA ALREADY IN HAND (DAYS)

- **CanadaBuys + SAM.gov + LACity**: map the fields already sitting in the feed/RawJson — contact email, commodity code→discipline, city, value, attachment URLs. No new scraping. **quick**

TIER 1 — THE ENABLING MODEL CHANGE (UNBLOCKS EVERYTHING)

- Add **Discipline, BuyerContact, and Documents** slots to OpportunityCandidate; stop hardcoding Discipline = Unknown in IngestionService — derive it from commodity/GSIN/NAICS codes (a prime-discipline classifier already exists as a model). **core**

TIER 2 — GENERALIZE THE APC JOB TO A LIVE-OPP DETAIL ENRICHER (THE REAL FIX)

- One decoupled enrichment job, modelled on APC, that opens the detail page for live BC Bid / Bids&Tenders / MERX / Bonfire opps and harvests scope + discipline + contact + **documents** + plan-takers. Wire in the two probe tools that already do the login/read. **the load-bearing rung**
- Fix the plan-taker re-queue bug: opps with no supplier list get re-selected every run forever, starving the batch. **same pass**

TIER 3 — COVERAGE GAPS

- Add **Interior Health** (+ other health authorities) as Bids&Tenders tenants — today RIH-type RFPs aren't ingested at all.
- Surface Major Projects Inventory intel (value/owner/stage) onto the opportunity surface instead of a siloed table.
- Download RFP documents for live opps (reuse the historical document-download service).

What this audit is and isn't

- **Verified from source** (file:line for every claim): the listing-only behaviour of all six source families, the two-layer root cause, the APC template, and the DB outcome (discipline/value/contact ≈ 0 on live opps).
- **Not yet quantified:** exactly how many current live opps would flip from thin→rich per tier — that needs a backfill dry-run, which Tier 0/1 make cheap.
- **Deliberately deferred:** no code written. This is the audit you asked for first; the ladder above is a proposal awaiting your scope call.